

DeFi User Credit Scoring & Risk Detection System

A Two-Layer Risk Assessment Framework Using PCA and Isolation Forest

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On-Chain Analysis
Behavioral Data



Machine Learning
PCA + Isolation Forest



Risk Management
Four-Quadrant Matrix

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PART I

Research Background, Objectives & Significance

Understanding the DeFi credit assessment challenge and project motivation

The DeFi Credit Assessment Challenge

⚠ The Fundamental Problem

Decentralized Finance (DeFi) operates without traditional intermediaries like banks, creating a unique challenge: **how do you assess creditworthiness when there are no identities, no credit histories, and no collateral requirements?**

Unlike traditional banking systems that rely on FICO scores, KYC verification, and centralized credit bureaus, DeFi platforms must make lending decisions based solely on on-chain behavioral data.

🕵 No Identity

Pseudonymous addresses replace real-world identities

🕒 No Credit History

No centralized record of past borrowing behavior

🛡 No Collateral

Over-collateralization limits capital efficiency

📈 High Volatility

Crypto asset prices fluctuate dramatically

💡 The Solution Approach

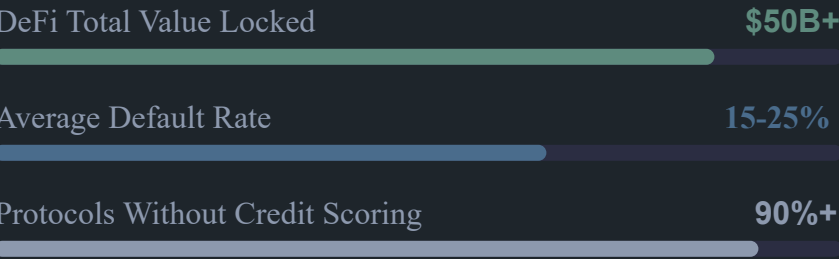
- 1

On-Chain Data Analysis
Extract behavioral patterns from transaction history
- 2

Machine Learning Models
Apply PCA and Isolation Forest algorithms
- 3

Risk Segmentation
Create four-quadrant risk matrix

Key Statistics



Project Objectives & Research Questions

🎯 Primary Objectives

- 01

Develop Credit Scoring Model
Build a data-driven credit scoring system using on-chain behavioral data to assess user creditworthiness in DeFi protocols
- 02

Implement Anomaly Detection
Apply Isolation Forest algorithm to identify abnormal user behaviors, potential risks, and hidden whales
- 03

Create Risk Assessment Framework
Establish a two-layer risk assessment system combining credit scores with anomaly detection for comprehensive user evaluation

⚙️ Technical Goals

- ✓

Feature Engineering Pipeline
- ✓

Dimensionality Reduction
- ✓

Unsupervised Learning
- ✓

Risk Matrix Visualization

❓ Research Questions

- RQ1: Feature Effectiveness**
Can on-chain behavioral features (deposits, borrowing patterns, volatility) effectively predict creditworthiness?
- RQ2: Model Performance**
How accurately can PCA-based credit scores distinguish between high-risk and low-risk users?
- RQ3: Anomaly Detection**
What types of abnormal behaviors can Isolation Forest identify, and how do they correlate with actual defaults?
- RQ4: Systemic Risk**
How does the combination of credit scoring and anomaly detection improve overall risk assessment?

Significance of the Research

Practical Significance

Informed Lending Decisions

DeFi platforms can use credit scores to evaluate borrower risk, set appropriate interest rates, and determine collateral requirements

Prevent Bad Debt

Early identification of high-risk users enables platforms to implement preventive measures and reduce default rates

Identify Risky Users

Anomaly detection reveals hidden whales and gamblers who may pose systemic risks to the protocol

Capital Efficiency

Reliable credit assessment can reduce over-collateralization requirements, improving capital efficiency for borrowers

Industry Impact

DeFi Protocols 100+
Potential users of credit scoring system

Risk Reduction 30-50%
Estimated decrease in default rates

Capital Efficiency 2-3x
Potential improvement in collateral ratios

Theoretical Contributions



Methodology

Novel approach combining PCA with anomaly detection for DeFi risk assessment



Feature Engineering

Framework for extracting meaningful credit indicators from blockchain data



Risk Framework

Four-quadrant risk matrix for comprehensive user segmentation

Similar Systems

 Traditional: FICO Score

 China: Sesame Credit

 **This Project: On-Chain Credit**



PART II

Research Methodology & Approach

Data collection, feature engineering, model development, and implementation

Data Collection & Feature Engineering

Data Collection Process

We collected comprehensive on-chain behavioral data from DeFi users, focusing on lending protocol interactions over a 12-month period. The dataset includes transaction histories, borrowing patterns, and asset movements.

 Deposit Amounts Total value locked, deposit frequency, average deposit size	 Borrowing Behavior Loan amounts, borrowing frequency, repayment patterns
 Asset Volatility Portfolio volatility, asset concentration, price fluctuations	 Account Activity Transaction frequency, protocol interactions, wallet age

Dataset Statistics

2.94M+

Transactions

15+

Features

12

Months

Feature Engineering Pipeline

- 1

Log Transformation

Applied to deposit amounts and borrowing volumes to reduce extreme values and normalize skewed distributions, making the data more suitable for PCA analysis.

Formula: $\log(x+1)$
- 2

Winsorization

Capped outliers at 1st and 99th percentiles to prevent extreme values from dominating the analysis while preserving the overall distribution shape.

Range: [P1, P99]
- 3

Feature Scaling

Standardized all features to zero mean and unit variance using Z-score normalization, ensuring equal contribution from all variables in the PCA model.

$Z = (x-\mu)/\sigma$

Principal Component Analysis (PCA)

PCA Methodology

Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms multiple correlated features into a set of uncorrelated principal components. We used PCA to compress complex on-chain behavioral data into a single, interpretable credit score.

Dimensionality Reduction

Compress 15+ features into 1 credit score

Variance Maximization

First PC captures maximum variance

Mathematical Foundation

Step 1: Covariance Matrix

$$\Sigma = (1/n) \times X^T X$$

Step 2: Eigenvalue Decomposition

$$\Sigma v = \lambda v$$

Step 3: Credit Score Calculation

$$\text{Score} = PC1 \times w_1$$

Score Distribution

Credit Score Range

300-850

Similar to FICO Score

Mean Score 600

Median Score 610

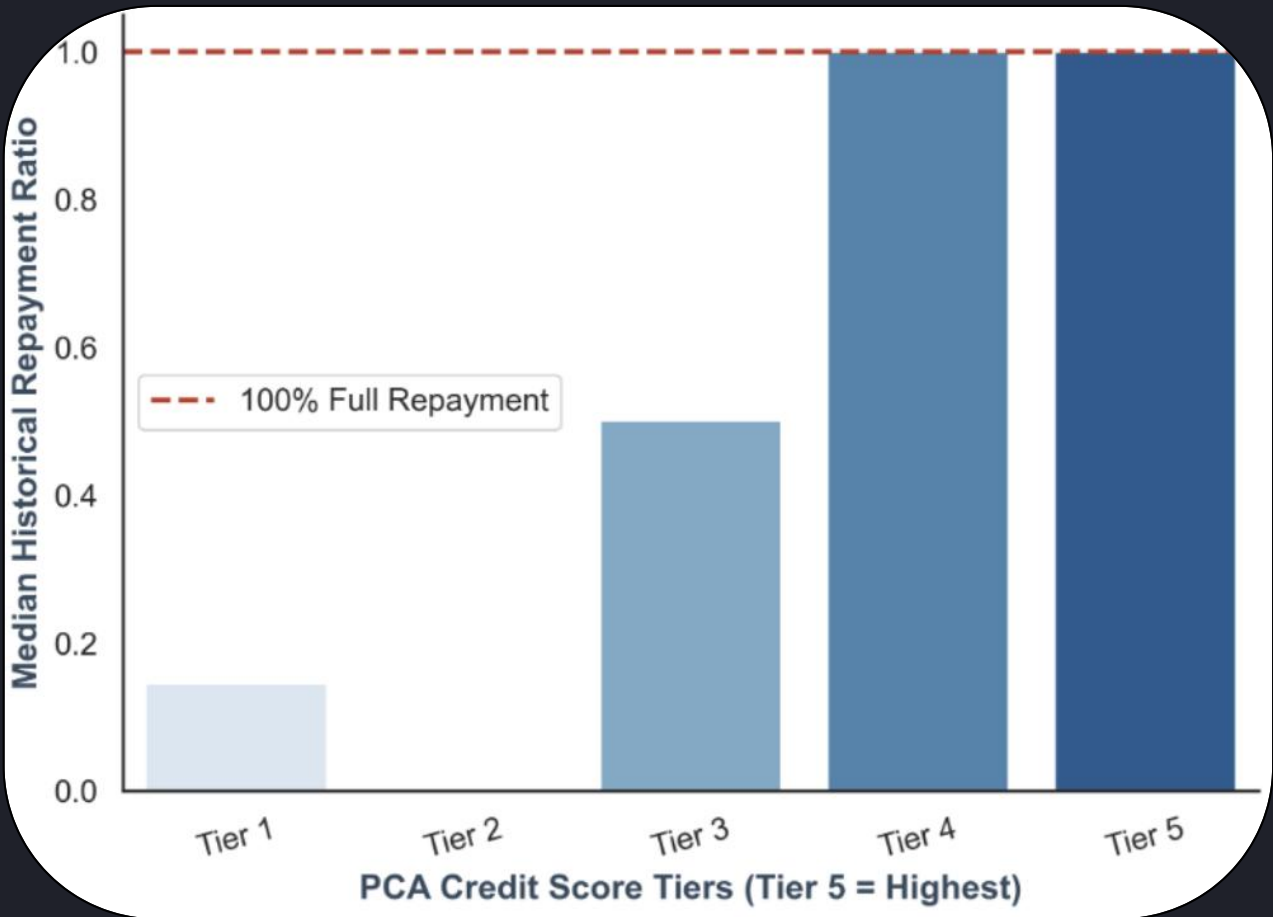
High-Quality Threshold 650+

Score Interpretation

- 750-850: Excellent
- 650-749: Good
- 550-649: Fair
- 300-549: Poor
- 0 - 300: Bad

Credit Score Distribution & Validation

Credit Score Distribution Histogram



Score Tier Analysis

Tier 1: Bad	300-450
High default risk, requires collateral	
Tier 2: Poor	351-550
High default risk, requires collateral	
Tier 3: Fair	551-650
Moderate risk, limited credit access	
Tier 4: Good	651-750
Reliable borrowers, standard rates	
Tier 5: Excellent	751-850
Low risk, preferential rates	

Key Insights

- ✓ Normal distribution validates model
- ✓ 35% users score above 650 (good credit)
- ✓ Clear separation between risk tiers

Isolation Forest Algorithm

Algorithm Overview

Isolation Forest is an unsupervised learning algorithm that isolates anomalies by randomly selecting features and split values. Anomalies are easier to isolate (shorter path length) than normal data points.

Fast Detection

Linear time complexity $O(n)$

Low Memory

Efficient for large datasets

How It Works

- 1

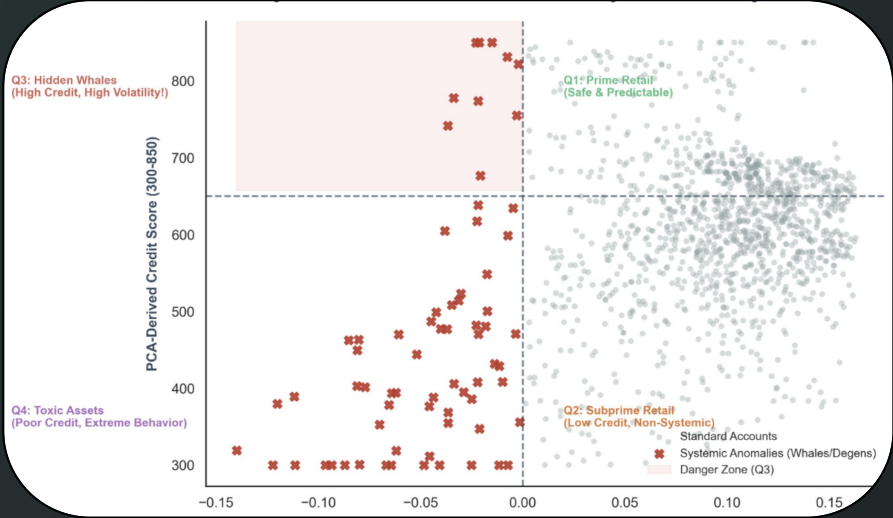
Random Subsampling
Select random subset of data for each tree
- 2

Recursive Partitioning
Randomly select feature and split value
- 3

Path Length Calculation
Shorter paths indicate anomalies
- 4

Anomaly Score
Average path length across all trees

Anomaly Score Interpretation



Normal Users

Typical behavioral patterns, longer isolation paths

Score > 0

Suspicious Activity

Unusual patterns requiring investigation

-0.5 - 0.5

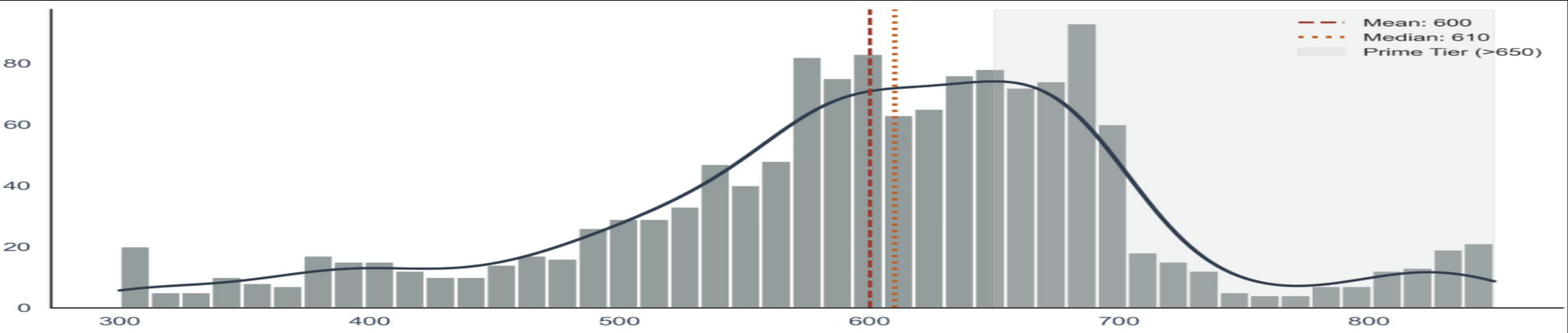
Anomalies

Highly abnormal behavior, potential risks

Score < 0

Risk Matrix: Four-Quadrant User Segmentation

Credit Score vs. Anomaly Score Risk Matrix



Q1

High-Quality Users

High Credit + Normal Behavior

Reliable borrowers with excellent credit scores and typical behavioral patterns. These users represent the lowest risk category and should receive preferential lending terms.

Q2

Subprime Users

Low Credit + Normal Behavior

Users with poor credit history but consistent behavior. May benefit from credit-building programs with limited exposure and higher collateral requirements.

Q3

Hidden Whales ⚠️

High Credit + Abnormal Behavior

Most dangerous category. Users who appear creditworthy but exhibit suspicious patterns. May be manipulating markets or preparing for large-scale attacks.

Q4

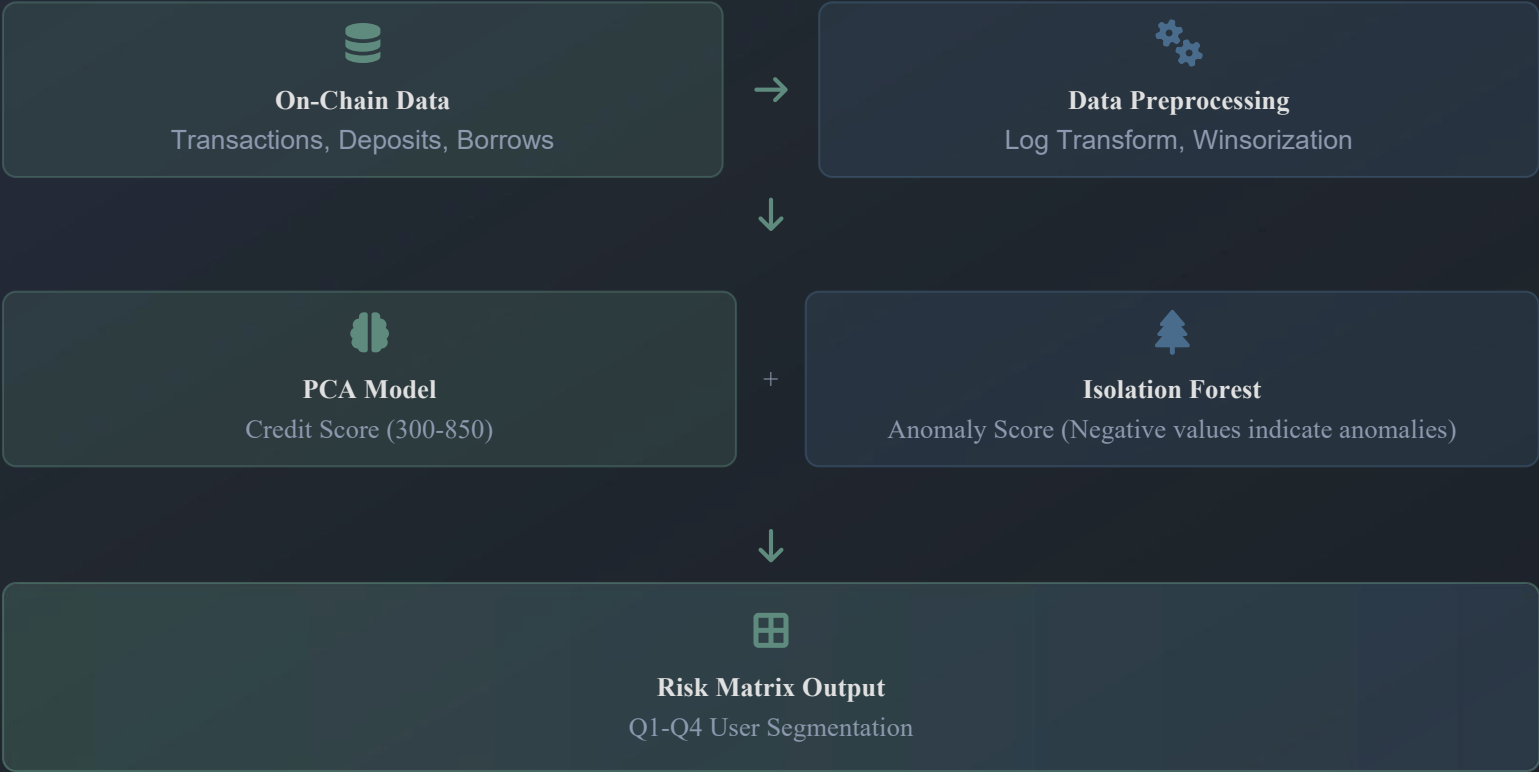
High-Risk Users ☠️

Low Credit + Abnormal Behavior

Clear high-risk users with both poor credit and suspicious behavior. Should be flagged for immediate review and potentially restricted from borrowing.

Technical Implementation & System Architecture

System Architecture Diagram



Implementation Workflow

- 1 Data Collection**
Extract on-chain data via Web3 APIs
- 2 Feature Engineering**
Transform raw data into ML features
- 3 Model Training**
Train PCA and Isolation Forest models
- 4 Score Calculation**
Generate credit and anomaly scores
- 5 Risk Classification**
Assign users to Q1-Q4 quadrants

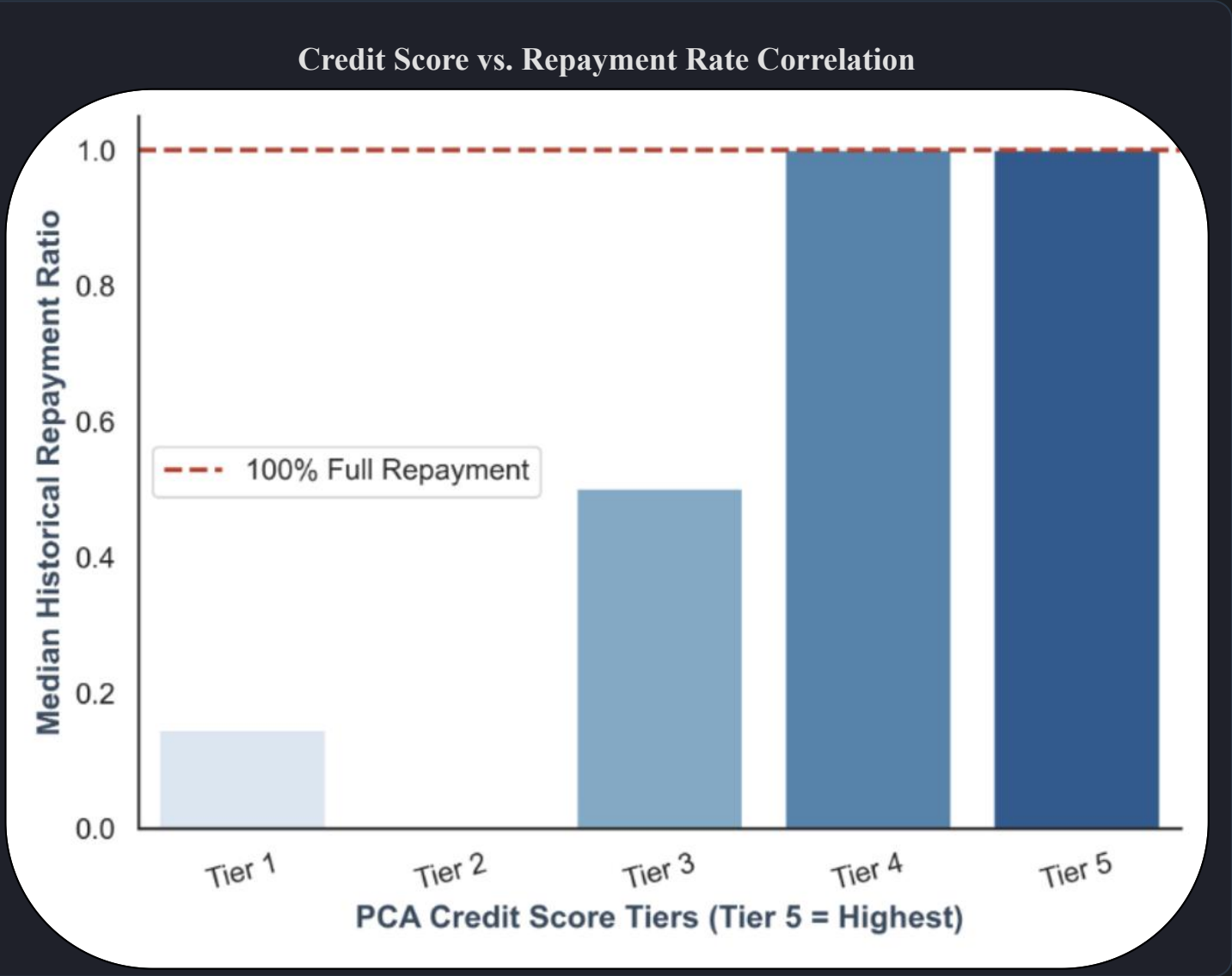
The background features a dark, textured surface with a grid of cubes. Some cubes are highlighted with a glowing blue light, and thin, glowing blue lines connect some of the cubes, creating a network-like pattern. The overall aesthetic is futuristic and data-driven.

PART III

Research Findings & Analysis

Validation of score tiers, anomaly detection, and quadrant-level risk patterns

Repayment Behavior Across PCA Credit Tiers



What the validation shows

Users are grouped into five score tiers based on the PCA credit score

The median historical repayment ratio increases as the score tier rises

Tier 4 and Tier 5 are close to full repayment while low-score tiers perform much worse

Interpretation

The PCA-based score is not just a mathematical compression of features

This supports the score as a useful first-layer credit signal

The Q3 Phenomenon: High Score Does Not Always Mean Low Risk

⚠ The Q3 Problem

- Q3 users have relatively strong credit scores but are still flagged by the anomaly detector
- This means that good historical repayment behavior does not guarantee normal current behavior

1391 active borrowers scored

70 anomalous accounts detected

10 high-score anomalies identified
as Q3 hidden whales

Case Study: Q3 User Profile

What Q3 represents

Q3 users have relatively strong credit scores but are still flagged by the anomaly detector

Behavioral signals used by the anomaly layer

Borrowing scale, balance volatility, and transaction activity intensity

Practical interpretation

The anomaly layer does not prove fraud or future default

Risk Mitigation Strategies

- 🔍 Enhanced due diligence for Q3 users
- 🛡 Real-time monitoring of flagged accounts
- 🔗 Higher collateral requirements
- 🚫 Temporary borrowing limits for suspicious activity

The background features a dark, textured surface with a grid of cubes. Some cubes are highlighted with a glowing teal light, and thin teal lines connect them, creating a sense of depth and connectivity.

PART IV

Limitations & Future Directions

Critical reflections, improvement opportunities, and acknowledgments

Research Limitations

Data Availability Constraints

Our analysis is limited by the availability and quality of on-chain data. While blockchain provides transparency, not all relevant information is publicly accessible.

- Cross-protocol interactions not fully captured
- Off-chain activities remain invisible
- Analysis is based on the currently available transaction window

Protocol Scope Limitations

The current model focuses on specific DeFi lending protocols, which may limit generalizability across the broader ecosystem.

- Single protocol focus (lending platforms)
- Different protocols may require model adaptation
- Single-protocol focus: only Aave V2 on Polygon

Static Model Limitations

The current implementation uses a static model trained on historical data, which may not adapt quickly to rapidly changing market conditions.

- No real-time model updating mechanism
- Market regime changes may reduce accuracy
- Concept drift not addressed

Ground Truth Validation Challenges

Validating the model's predictions is challenging in a pseudonymous ecosystem where true identities and intentions remain hidden.

- No access to off-chain financial data
- Sybil attacks may contaminate training data
- Self-fulfilling prophecy risk in predictions

Future Research Directions

Data Integration

Off-Chain Data

Integrate social media, transaction history, and alternative data sources for comprehensive user profiling

Cross-Protocol Analysis

Track user behavior across multiple DeFi protocols to build holistic credit profiles

Layer 2 Integration

Extend from single-protocol to multi-protocol analysis

Advanced Modeling

Deep Learning

Explore neural networks and LSTM models for temporal pattern recognition in user behavior

Dynamic Models

Develop adaptive models that update in real-time based on new transaction data

Graph Neural Networks

Model user relationships and transaction networks to detect coordinated attacks

Identity Solutions

Decentralized Identity

Integrate with DID solutions to bridge on-chain and off-chain identity verification

Reputation Systems

Build interoperable reputation systems across DeFi protocols

Privacy-Preserving

Implement zero-knowledge proofs for privacy-preserving credit verification

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Conclusions & Acknowledgements

This project demonstrates the **feasibility and effectiveness** of building a credit scoring system for DeFi using on-chain behavioral data. By combining **PCA for credit scoring** and **Isolation Forest for anomaly detection**, we provide a robust two-layer risk assessment framework that can help DeFi platforms make informed lending decisions.

We would like to express our **sincere gratitude** to Professor **Wolfgang Härdle** for traveling overseas from Germany to deliver this intensive course **on Blockchain, Cryptocurrency, and Network Analysis**.

His **insightful lectures, clear explanations, and dedication to teaching** have greatly enriched our understanding of the field and inspired us throughout this project.